Problem Set for 09 March 2025

Definition For any linear map $f : U \to V$,

1. $\ker(f) = \mathrm{Null}(f) = N(f) = f^{-1}(0) = \{u \in U \mid f(u) = 0\}$ is a subspace of U;
2. $\mathrm{im}(f) = \mathrm{Range}(f) = R(f) = f(U) = \{f(u) \mid u \in U\}$ is a subspace of V.

Exercise 1 Let $V = \mathbb{C}[x]$, and let $\omega = \frac{-1+\sqrt{3}i}{2}$ denote a cubic root of unity ($\omega^3 = 1$).

1. Prove that each $V_k := \{f \in \mathbb{C}[x] \mid f(x) = \omega^k \cdot f(\omega x)\}$ is a linear subspace of V.
2. Prove that $V = V_0 \oplus V_1 \oplus V_2$.
3. Prove that the mapping

$$L : V \to V, \quad f(x) \mapsto \omega \cdot f(\omega x) - \omega^2 f(\omega^2 x)$$

   is indeed a linear transformation.
4. Compute $\mathbf{im}(L)$ and $\mathbf{ker}(L)$.
5. Say $(u, \lambda) \in V \times \mathbb{C}$ is an eigenpair of L, provided $u \neq \mathbf{0}$ and $Lu = \lambda u$. Determine all eigenpairs of L.

> *Hint: find $A \in \mathbb{C}^{3\times 3}$ such that*
>
> $$(L(f(x)) \mid L(f(\omega x)) \mid L(f(\omega^2 x))) = (f(x) \mid f(\omega x) \mid f(\omega^2 x)) \cdot A.$$

Exercise 2 (blank-filling questions) Provide explicit examples for each of the following cases **WITHOUT** verification:

1. A linear endomorphism $f : V \to V$ that is injective but not surjective.
2. A linear endomorphism $g : V \to V$ that is surjective but not injective.
3. A linear endomorphism $h : V \to V$ over the $\mathbb{C}$-vector space V such that for every $z \in \mathbb{C}$, the equation $h(v) = z \cdot v$ has only the trivial solution $v = 0$. In other words, h has no eigenvalues.
4. A linear endomorphism $f : V \to V$ that is injective but not surjective, a linear endomorphism $g : V \to V$ that is surjective but not injective, and such that $g \circ f = \mathrm{id}_V$.
5. **(Optional)** A linear endomorphism $f : V \to V$ that is injective but not surjective, a linear endomorphism $g : V \to V$ that is surjective but not injective, and such that $f + g$ is an isomorphism (a linear bijection).